AI PRACTICE DEBUGGING REPLAY

# Debug a Queue Backlog - Production Debugging Replay

A 31-minute incident conversation for diagnosing rates, age, retries, dead letters, partition skew, downstream limits, safe containment, recovery, and prevention.

Protect the user promise, measure effective arrivals and successful service, stop amplification, and scale only when parallelism and downstream headroom are proven.

VERSION 1.0

FORMAT PRODUCTION DEBUGGING REPLAY + RETEST

STATUS LEARNER READY

AI-PRACTICE / QUEUE-BACKLOG-DEBUGGING-REPLAY

PROTECT THE PROMISE BEFORE TOUCHING CAPACITY

# A queue backlog is a symptom, not a diagnosis

The first job is to identify affected work, oldest age, arrival pressure, successful service rate, retry amplification, partition skew, and downstream health. Adding workers is a conditional intervention, not a default answer.

**TEACHING-DATA BOUNDARY**

**This is a scripted teaching replay. Names, rates, scores, incidents, timestamps, and outcomes are fictional sample data and do not represent a learner or employer. Adapt the method and use only verified production evidence.**

| 01 | 02 | 03 | 04 | 05 |
|---|---|---|---|---|
| **Impact** | **Rates** | **Cause** | **Contain** | **Recover** |
| Name the user promise, affected priority, start time, and accepted-work risk. | Separate new arrivals, retry arrivals, successful completions, and backlog age. | Test demand, consumers, partitions, poison work, and downstream dependencies. | Protect critical traffic and stop amplification before expanding concurrency. | Drain safely, reconcile effects, prove age falls, and prevent recurrence. |

## The 31-minute incident conversation

| Clock | Decision | Minimum output |
|---|---|---|
| 0-3 | Define impact, priority, onset, and data-loss/duplicate risk. | One user promise and one protection goal. |
| 3-8 | Read age, arrivals, successful completions, retries, skew, and dependency health. | A rate model plus the most dangerous unknown. |
| 8-13 | Rank hypotheses and choose fast confirming/disconfirming checks. | Three hypotheses ordered by evidence and reversibility. |
| 13-18 | Contain user harm and retry amplification. | One protection action, owner, and rollback trigger. |
| 18-23 | Bound retries, quarantine poison work, and define safe re-drive. | Attempt/deadline budget plus DLQ ownership. |
| 23-28 | Choose drain or scale conditions and verify downstream headroom. | Conditional capacity decision and approximate drain range. |
| 28-31 | Declare recovery, reconcile, communicate, and prevent recurrence. | Exit criteria and the next durable guardrail. |

## Say this before proposing a fix

- ☐ Which workflow is delayed, what deadline or user promise is at risk, and which priority/tenant/partition is worst?
- ☐ Did accepted work stop completing, complete twice, expire, or move into an unowned dead-letter path?
- ☐ What changed near the onset: demand, deploy, configuration, schema, dependency, quota, or partition ownership?
- ☐ Can optional producers be paused or critical work isolated without violating a stronger contract?

USE RATES AND AGE TO DESCRIBE THE INCIDENT

# Depth alone can improve while users get worse

Measure new work, retry work, successful effects, and oldest age separately. A falling queue depth may reflect expiry, drops, or dead-letter movement rather than recovery.

**RATE MODEL**

**Backlog change per second is approximately new arrivals + retry arrivals - successful completions. When successful completions exceed effective arrivals and rates remain stable, drain time is approximately backlog divided by the net drain rate.**

FICTIONAL WORKED EXAMPLE - ADAPT THE METHOD, NOT THE FACTS

| Moment | Observed rates | Meaning | Safe inference |
|---|---|---|---|
| Before containment | 500 new/s + 750 retry/s; 400 successful/s | Backlog grows about 850/s. Retries exceed new demand. | Stop amplification and inspect dependency errors before adding concurrency. |
| After retry bound | 200 effective arrivals/s; 800 successful/s | The theoretical net drain is about 600/s. | Estimate a drain range only if partitions, priorities, quotas, and rates remain stable. |
| Depth falls, oldest age rises | Low-priority depth drops; critical oldest age reaches 22 min | Global depth hides starvation or expiry in a critical lane. | Segment age and successful throughput by priority, tenant, and partition. |
| Attempts rise, success is flat | 1,600 dequeues/s; 420 successful effects/s | More work is being attempted, not completed. | Track acknowledgements and effect completion separately from dequeue rate. |

## The minimum incident panel

| Signal | Why it matters |
|---|---|
| Oldest age by priority/tenant/partition | Directly exposes deadline risk, starvation, and localized hot spots. |
| New arrivals and retry arrivals | Separates demand from incident-created amplification. |
| Successful completions | Measures useful service; attempts and dequeues can rise while effects fail. |
| Processing time and downstream latency/errors/quota | Shows whether consumers or a dependency set the real service ceiling. |
| Partition lag, ownership, and saturation | Reveals skew, idle workers, unavailable leases, and misleading global capacity. |
| DLQ ingress, reason, oldest age, and re-drive status | Makes quarantined work visible and accountable rather than discarded. |

TEST A CAUSAL TREE, NOT A FAVORITE EXPLANATION

# Every hypothesis needs confirming and disconfirming evidence

Start with the cheapest safe checks that separate demand, service degradation, skew, poison work, retry storms, and downstream saturation. Preserve evidence before changing the system.

| Hypothesis | Confirming evidence | Disconfirming evidence | First safe check |
|---|---|---|---|
| Demand spike | New-arrival rate rises before age and backlog. | New arrivals are normal while retries surge. | Compare producer accepts by tenant/event type with the prior baseline. |
| Consumer regression | Processing time or failure starts with a deploy/config change. | Old and new versions have the same profile. | Split success/latency by version and canary a rollback if safe. |
| Hot partition | One partition ages while workers elsewhere are idle. | Lag and ownership are balanced. | Inspect key distribution, leases, and partition ownership. |
| Poison message | Same item/error repeats and blocks ordered work. | Failures are broad and input-independent. | Trace one safe identifier and compare the next item in that key. |
| Dependency throttle | Remote latency, 429/5xx, quota, or pool saturation rises first. | Dependency headroom and latency remain normal. | Compare successful effects with dependency capacity and error classes. |
| Retry storm | Retry arrivals and synchronized attempts dominate new work. | Attempts are mostly first tries. | Group attempts by original operation, error class, and retry age. |

INCIDENT DISCIPLINE

### Preserve evidence

Record onset, deploy/config identifiers, representative failure classes, rates, age, and partition state before broad restarts or purges erase the sequence.

INCIDENT DISCIPLINE

### Prefer reversible moves

Pause optional producers, reduce retry concurrency, isolate priority, roll back a suspect change, or canary capacity with explicit stop conditions.

NEVER OPTIMIZE THE CHART

**Do not purge, expire, or mass-dead-letter accepted work merely to make depth fall. Any discard or replay requires an authorized business policy, impact assessment, and auditable decision.**

CONTAIN BEFORE YOU SCALE

# More workers can accelerate the failure

Concurrency helps only when work is parallelizable, partitions or leases are available, and every downstream limit has headroom. Otherwise it increases timeouts, lock pressure, database connections, duplicate effects, or provider throttling.

| 01 | 02 | 03 | 04 | 05 |
|---|---|---|---|---|
| **Protect** | **Reduce** | **Isolate** | **Canary** | **Reconcile** |
| Reserve or prioritize capacity for deadline-critical and transactional work. | Pause optional producers and bound new retry creation. | Circuit-break failing dependencies or quarantine poison work by policy. | Change concurrency or rollout on one bounded slice with stop signals. | Prove accepted effects, duplicates, skips, and dead letters are accounted for. |

| Intervention | Use when | Do not use when | Stop/rollback signal |
|---|---|---|---|
| Add workers | Partitions and downstream capacity are available; CPU or worker slots are the bottleneck. | Dependency quota, locks, DB pool, poison work, or retry traffic sets the ceiling. | Downstream errors, saturation, duplicates, or oldest age worsen. |
| Pause optional producers | New low-priority demand threatens a stronger accepted-work promise. | The paused work is itself the highest-priority contract or cannot be recovered. | Critical age does not improve or lost-input risk exceeds policy. |
| Reduce retry concurrency | Transient failures are amplifying load faster than new work. | A deadline-critical retry would be silently abandoned. | Successful throughput falls without reducing dependency pressure. |
| Quarantine poison work | A permanent or malformed item blocks ordered or shared progress. | Skipping violates strict per-key ordering or an authorized loss policy. | Ordering/effect integrity cannot be preserved for affected keys. |

## Recovery is a set of exit conditions

+ Successful throughput exceeds effective arrivals with margin across critical partitions.
+ Oldest age falls for the affected priority and tenant, not only globally.
+ Downstream latency, errors, quotas, pools, and locks remain inside safe operating bounds.
+ Retries return to baseline, DLQ ingress stops, and quarantined work has an owner and plan.
+ Accepted effects are reconciled for missing, duplicate, expired, or out-of-order outcomes.

TREAT RETRIES AND DEAD LETTERS AS PRODUCT BEHAVIOR

# Bound attempts, time, side effects, and re-drive

Retries are new load. A dead-letter queue is a controlled evidence and recovery path, not a trash bin and not proof that work is correct.

| Failure class | Retry policy | Required evidence |
|---|---|---|
| Transient and safe | Exponential backoff with jitter, attempt and elapsed-time budget, deadline awareness. | Original operation ID, attempt number, error class, first/last failure time. |
| Permanent input error | Do not repeat unchanged work; quarantine under an explicit product policy. | Validation reason, safe reference, owning team, customer-visible consequence. |
| Ambiguous effect | Check an idempotency/effect ledger or reconcile before repeating the external action. | Stable idempotency key, prior response/effect state, reconciliation result. |
| Dependency overload | Respect server guidance, reduce concurrency, randomize retries, and stop at the user deadline. | Quota/headroom, retry-after signal where supplied, effective retry arrival rate. |

## A safe dead-letter record

- [ ] Stable operation reference and safe payload pointer; redact secrets and unnecessary personal data.
- [ ] Failure class, first and last failure timestamps, attempt count, producer/consumer version, and partition/key context.
- [ ] Business priority, expiry or ordering consequence, current owner, alert state, and retention policy.
- [ ] Re-drive requires an approved fix, effect-level idempotency, a canary batch, rate limit, monitoring, pause control, and audit trail.

DECISION

### Ordering trade-off

Skipping a poison item can restore throughput but violate strict per-key order. State whether the product prefers blocking, quarantine with key isolation, or an explicit compensating path.

DECISION

### Visibility trade-off

A short visibility/ack deadline can duplicate slow work. A very long deadline delays takeover after failure. Use extension/heartbeat only with bounded ownership.

IDEMPOTENCY BOUNDARY

**Broker deduplication does not guarantee the external effect occurs once. Use a stable operation key and record or reconcile the effect at the system that owns the business consequence.**

REPLAY THE WEAK ANSWER WITHOUT DISCARDING ITS USEFUL INSTINCT

# Capacity may help - after rates, age, retries, and headroom are known

The scripted response begins with adding workers and monitoring queue length. Keep the valid capacity hypothesis, then insert the missing diagnostic and safety decisions before acting.

FICTIONAL WORKED EXAMPLE - ADAPT THE METHOD, NOT THE FACTS

| Interviewer probe | Weak-but-useful move | Missing decision | Improved response |
|---|---|---|---|
| The queue is growing. What first? | Add more workers and watch queue length. | Impact, oldest age, effective arrivals, successful service, partitions, and downstream headroom. | I will first protect critical work and compare oldest age, new/retry arrivals, successful completions, partition skew, and downstream capacity; workers are conditional. |
| Why not scale immediately? | More consumers should drain faster. | Parallelism and the true service ceiling. | I will canary concurrency only if partitions are available and downstream latency, quotas, pools, and locks have headroom. |
| What enters the DLQ? | Messages that fail many times. | Failure class, deadline, effect ambiguity, ordering, ownership, and re-drive safety. | Permanent invalid work is quarantined with evidence; ambiguous effects are reconciled; re-drive is canaried, idempotent, rate-limited, owned, and audited. |
| How do you know recovery is real? | Queue depth returns to normal. | Depth can fall through expiry, drops, or starvation. | Recovery requires successful throughput above effective arrivals, falling oldest age across critical partitions, safe dependencies, bounded retries, and reconciled effects. |

## Common incident-answer mistakes

+ Using dequeue or attempt rate as successful service rate.
+ Applying a global drain-time estimate while one priority or partition remains blocked.
+ Treating backoff without jitter as sufficient protection against synchronized retries.
+ Moving failed work to a DLQ without owner, alert, retention, ordering policy, or re-drive control.
+ Scaling consumers while the database, provider quota, lock, or connection pool is already saturated.
+ Calling the incident resolved before accepted effects are reconciled.

EXPLAIN THE 71 AND RUN A COMPARABLE CORRECTION

# A score is useful only when it changes the next observable behavior

This representative catalog score is scripted teaching data. It identifies why retry bounds and recovery evidence arrived too late; it is not account history or a production incident assessment.

FICTIONAL SCRIPTED SCORECARD - ADAPT THE RUBRIC, NOT THE SCORE

| Dimension | Score | Evidence | Correction |
|---|---|---|---|
| Problem framing | 72 | The backlog was recognized, but user promise and oldest-age segmentation were not first. | Open with impact, priority, accepted-work risk, and the rate model. |
| Technical depth | 74 | Workers and queue monitoring were plausible, but retries, DLQ, idempotency, and partitions lacked bounds. | Classify failures and state attempt, deadline, effect, ordering, and re-drive rules. |
| Trade-off clarity | 62 | Capacity was presented as a default rather than a conditional intervention. | Name when workers help, when they amplify collapse, and the stop signal. |
| Communication | 76 | The response was direct but skipped the evidence sequence required for a safe incident decision. | Use impact -> rates -> hypothesis -> containment -> recovery -> prevention. |

CORRECTED OPENING

**I will first protect the affected deadline and establish oldest age, new and retry arrivals, successful completions, partition skew, and downstream headroom. I will bound retry amplification and isolate poison work before adding workers. I will scale only if parallelism and dependencies allow it, and declare recovery only when successful throughput exceeds effective arrivals and oldest age falls across critical partitions.**

| Practice block | Action | Success evidence |
|---|---|---|
| Replay - 6 min | Mark every claim as observation, hypothesis, action, or exit condition. | No mechanism is presented as evidence of recovery. |
| Rate model - 5 min | Calculate growth/drain ranges and list conditions that make them invalid. | New, retry, and successful rates remain separate. |
| Pressure - 8 min | Answer worker, DLQ, priority, poison, idempotency, and re-drive probes. | Each answer includes a boundary and stop condition. |
| Transfer - 9 min | Debug a nearby file-scan or notification backlog under the same clock and rubric. | The method transfers without copying the original fix. |
| Close - 3 min | Select one prevention guardrail and accountable owner. | Alert threshold, action, and review date are explicit. |

## Completion checklist

- ☐ Impact and oldest age are segmented by priority, tenant, and partition.
- ☐ New arrivals, retry arrivals, successful completions, and downstream ceiling are separate.
- ☐ The chosen containment is reversible and has an owner plus rollback signal.
- ☐ Retries, dead letters, effect idempotency, ordering, and re-drive are bounded.
- ☐ Recovery includes falling age and reconciliation, not merely lower depth.
- ☐ The re-test uses comparable conditions and records transcript evidence for the score.

NEXT ACTION

**Rehearse the corrected opening once, then run the same six pressure probes on an adjacent backlog scenario. Stop when the evidence sequence is audible without prompting.**